

ECO362 Final(2022) Solutions

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Q1

- (a) F: If we take CAPM as true, a stock with negative beta can hedge systemic risk.
- (b) F: Idiosyncratic risk does not command extra expected return if APT holds (alpha is zero)
- (c) F: Index funds have been gaining market share in the past 15 years.
- (d) T: Using the 3-factor Fama-French for constructing our benchmarks, the information ratio can be used for performance evaluation independent of the sign of the betas of the benchmark portfolios(HML,SMB) for the two funds.
- (e) F: Life insurance could still reduce the variance of next-period wealth, so as long as you are risk-averse, the optimal amount of life insurance is between zero and human capital value.
- (f) F: According to the Black-Scholes model, all options of a same underlying and same maturity should have the same implied volatility.
- (g) T: it is suboptimal for tax purpose. Tax-exempt municipal bonds should be held in a taxable brokerage account. Treasury bonds should be held in a 401(k) account. In this way, one doesn't need to pay tax for either position.

Q2

- (a) Stocks with the lowest price-dividend ratios are value stocks. Thus, Exxon Mobil and Johnson & Johnson are value stocks.

Grading:

4 points for two correct stocks.

2 points for one correct stock.

- (b) Stocks with the highest price-dividend ratios are growth stocks. Thus, Apple and Microsoft are growth stocks.

Grading:

4 points for two correct stocks.

2 points for one correct stock.

- (c) The lower price-dividend ratio means the higher expected return, given the same expected dividend growth rate. Thus, Exxon Mobil has the higher expected return.

Grading:

4 points for the correct conclusion and explanation.

2 points for the correct stock.

2 points for the correct explanation.

- (d) First, we can calculate the expected return of Microsoft, $\bar{R}_{MS}$, and use it to calculate the expected dividend growth of Apple, G_{Apple} . According to the Gordon growth model,

$$\bar{R}_{MS} = \frac{1 + G_{MS}}{P_{MS}/D_{MS}} + G_{MS} = \frac{1 + 5\%}{93} + 5\% = 6.129\%$$

$$G_{Apple} = \frac{(P_{Apple}/D_{Apple}) \cdot \bar{R}_{Apple} - 1}{1 + P_{Apple}/D_{Apple}} = \frac{167 * 6.129\% - 1}{1 + 167} = 5.497\%$$

Grading:

4 points for the correct equations as well as numerically correct $\bar{R}_{MS}$ and G_{Apple} .

2 points for the numerically correct values ($\bar{R}_{MS}$ and G_{Apple}).

2 points for writing the correct equations for both $\bar{R}_{MS}$ and G_{Apple} .

1 point for one numerically correct answer for either $\bar{R}_{MS}$ or G_{Apple} .

1 point for one correct equation for either $\bar{R}_{MS}$ or G_{Apple} .

- (e) If we assume no arbitrage, then we can say that the expected return is the fair compensation for the risk the stock has. It implies that the higher the expected return is, the higher the risk of the stock is. If we calculate the expected returns of Apple and Walmart using the Gordon growth model,

$$\bar{R}_{Apple} = \frac{1 + G_{Apple}}{P_{Apple}/D_{Apple}} + G_{Apple} = \frac{1 + 7\%}{167} + 7\% = 7.641\%$$

$$\bar{R}_{Walmart} = \frac{1 + G_{Walmart}}{P_{Walmart}/D_{Walmart}} + G_{Walmart} = \frac{1 + 5\%}{63} + 5\% = 6.667\%$$

Since $\bar{R}_{Apple}$ is higher than $\bar{R}_{Walmart}$, we can say that Apple is the riskier stock.

Grading:

4 points for the correct answer and explanation.

2 points for the correct explanation.

1 point for choosing the correct stock.

1 point for the correct calculation.

Q3

- (a) Actively managed funds charge management fees that corrode gross returns from investment.
- (b) No. It could be a result of luck that the fund happens to hold winner stocks and continues to perform better due to momentum.
- (c) These funds employ the small-cap growth strategy. Their SMB betas are positive, meaning that the portfolio tilts towards small-cap stocks. Their HML betas are negative, meaning that the portfolio tilts towards low book-to-market ratio stocks (i.e., growth stocks).

Grading:

4 pts: Correct conclusion and reasoning based on betas.

2 pts: Correct conclusion, or correctly recognize positive SMB betas and negative HML betas.

- (d) Using the lowest four portfolios, one can write the following four equations to solve for the four unknown factors:

$$\begin{cases} 0.01 = -0.40 + 0.93\overline{Market} + 0.32\overline{SMB} - 0.08\overline{HML} - 0.09\overline{WML} \\ 0.23 = -0.20 + 0.93\overline{Market} + 0.22\overline{SMB} - 0.10\overline{HML} - 0.02\overline{WML} \\ 0.34 = -0.13 + 0.93\overline{Market} + 0.20\overline{SMB} - 0.06\overline{HML} + 0.01\overline{WML} \\ 0.36 = -0.14 + 0.90\overline{Market} + 0.21\overline{SMB} - 0.03\overline{HML} + 0.04\overline{WML}. \end{cases} \quad (1)$$

In fact, one can use any four portfolios to write down equations like above.

Grading:

4 pts: Correct system of equations.

2 pts: A correct algebraic equation of the four-factor model without numerical substitution for known variables (mean excess returns, alphas and betas).

Q4

- (a) $\bar{y}$ represents the average short term riskless rate that the Fed sets. We solve for $\bar{y}$ and μ by plugging $y(1), y(5), y(10)$, and ρ into the term structure equation.

$$\begin{aligned}y(5) &= \bar{y} + \frac{1}{5} \left(\frac{1 - \rho^5}{1 - \rho} \right) (y(1) - \bar{y}) + \frac{4}{5} \mu \\y(10) &= \bar{y} + \frac{1}{10} \left(\frac{1 - \rho^{10}}{1 - \rho} \right) (y(1) - \bar{y}) + \frac{9}{10} \mu \\0.0387 &= \bar{y} + \frac{1}{5} \left(\frac{1 - (0.4)^5}{1 - 0.4} \right) (0.0457 - \bar{y}) + \frac{4}{5} \mu \\0.0366 &= \bar{y} + \frac{1}{10} \left(\frac{1 - (0.4)^{10}}{1 - 0.4} \right) (0.0457 - \bar{y}) + \frac{9}{10} \mu\end{aligned}$$

Solving the two equations gives us $\mu = 0.00415, \bar{y} = 0.03029$.

- (b) μ represents the term premium.
- (c) The 1-year rate is expected to fall. Students may justify this response by noticing that $y(1) > \bar{y}$, or by noticing that the yield curve slopes downward.
- (d)

$$\begin{aligned}y(20) &= \bar{y} + \frac{1}{20} \left(\frac{1 - \rho^{20}}{1 - \rho} \right) (y(1) - \bar{y}) + \frac{19}{20} \mu \\&= 0.03552\end{aligned}$$

Grading:

- 4 points for correct equation and consistent calculation. We give 4 points as long as the student's final numerical answer is consistent with the μ and $\bar{y}$ they computed in part (a).
- 3 points for correct equation but inconsistent calculation, e.g. due to numerical error. We give 3 points if the student reports a final numerical answer that is not consistent with the μ and $\bar{y}$ they computed in part (a).
- 2 points for correct equation that specifies $t = 20$ but no numerical answer.

We give no credit to any answer that provides a numerical answer without any equation or derivation.

Q5

- (a) Because we assume zero inflation risk premium, breakeven inflation is equal to expected annual inflation. Therefore, the 2-year expected annual inflation is $4.35\% - 2.00\% = 2.35\%$, and the 5-year expected annual inflation is $3.82\% - 1.48\% = 2.34\%$. Because the 2-year expected inflation is higher than the 5-year expected inflation, the market thinks the expected annual inflation will fall (slightly) over the next 5 years.

Grading:

Grades are awarded in increments based on the following criteria.

- (i) +1 point for saying breakeven inflation equals expected annual inflation when IRP is zero.
- (ii) +1 point for correctly computing the 2-year expected inflation.
- (iii) +1 point for correctly computing the 5-year expected inflation.
- (iv) +1 point for saying expected inflation will fall over the next 5 years based on the comparison between 2-year and 5-year expected inflations.
- (v) We deduct 0.5 point for each numerical mistake.

Additional notes for item (iv):

- We give credit as long as the answer's conclusion is consistent with the 2-year and 5-year breakeven inflation it computes: i.e. numerical mistakes will not be penalized twice.
- We give credit to answers that say expected inflation will stay flat because 2.35% and 2.34% are basically the same.
- We do not give credit to answers that reach a conclusion without calculating 2-year and 5-year breakeven inflation.

We deduct 1 point for any answer that does not compute 2-year and 5-year breakeven inflation, but only verbally states that "the 2-year breakeven inflation is higher than 5-year breakeven inflation".

- (b) Because we assume zero inflation risk premium, breakeven inflation is equal to expected annual inflation. Therefore, the 5-year expected annual inflation is $3.82\% - 1.48\% = 2.34\%$, and the 10-year expected annual inflation is $3.68\% - 1.49\% = 2.19\%$. Because the 5-year expected inflation is higher than the 10-year expected inflation, the market thinks the expected annual inflation will fall over the next 10 years.

Grading:

Grades are awarded in increments based on the following criteria.

- (i) +1 point for correctly computing the 5-year breakeven inflation.
- (ii) +1 point for correctly computing the 10-year breakeven inflation.
- (iii) +2 point for saying expected inflation will fall over the next 10 years based on the comparison between 5-year and 10-year expected inflations.
- (iv) We deduct 0.5 point for each numerical mistake. Carry-over error from part (a) for 5-year breakeven inflation is not penalized.

Additional notes for item (iii):

- We give credit as long as the answer's conclusion is consistent with the 5-year and 10-year breakeven inflation it computes: i.e. numerical mistakes will not be penalized twice.
- We do not give credit to answers that reach a conclusion without calculating 5-year and 10-year breakeven inflation.

We deduct 1 point for any answer that does not compute 5-year and 10-year breakeven inflation, but only verbally states that "the 5-year breakeven inflation is higher than 10-year breakeven inflation".

- (c) Inflation Risk Premium is the difference between expected inflation and breakeven inflation, where breakeven inflation is the difference between nominal and real treasury yields.

Inflation Risk Premium is driven by the covariance between inflation and Stochastic Discount Factor. The IRP is positive (negative) if inflation is positively (negatively) correlated with SDF.

Detailed Explanation: High SDF indicates bad states of the world. If inflation is positively correlated with SDF, inflation tends to be high in bad states of the world, thus the real payoff for investors in nominal bonds tends to be low in bad states of the world. In this case, nominal bonds must offer an extra risk premium, so IRP is positive.

Grading:

Grades are awarded in increments based on the following criteria.

- (i) +0.5 point for saying IRP is the difference between breakeven and expected inflation.
- (ii) +1 point for mentioning SDF or systematic risk in explaining how IRP can be positive or negative.
- (iii) +2.5 point for correctly saying that IRP is positive (negative) if inflation is positively (negatively) correlated with SDF.

We deduct 1 point if an answer says "IRP is positive (negative) if IRP is positively (negatively) correlated with SDF". This statement is inaccurate, because IRP is a number, not a random variable.

Additional notes:

- The part labelled "detailed explanation" above is for reference only, not required for full credit.
- If an answer does not mention SDF but otherwise provides the correct reasoning about how IRP is driven by covariance between inflation and state of the world / aggregate state of the economy, we give full credit for items (ii) and (iii).

- (d) Inflation risk premium equals breakeven inflation minus expected inflation. 5-year breakeven inflation is 2.34%. If 5-year expected annual inflation is 4%, then the inflation risk premium is $2.34\% - 4\% = -1.66\%$.

Grading:

- 4 points for correct formula and answer.

- 2.5 points for correct formula but wrong or no numerical answer, including any answer that neglects the percentage symbol in calculation.
- We do not penalize carry-over error for 5-year breakeven inflation.
- If an answer follows page 30 of the "Bonds" lecture and writes $\frac{1}{T} \ln(\text{IRP}) = -1.66\%$, and gets to the answer of $\text{IRP} = 0.9204$, give full credit.
- If an answer writes the difference between breakeven and expected inflation as $\frac{1}{T}\text{IRP}$ and gets $\text{IRP} = -1.66\% \times 5 = -8.30\%$, give full credit.

Q6

The calculations will be for a base interest rate swap of notional amount of 100 dollar. Then, we multiply everything by 20 million.

(a)

Calculate the annually compounded interest:

$$R_{f,2004} = \frac{100}{97.27} - 1 = 2.81\%$$

Grading:

3 points for the correct answer and formula

2 points for numerical error.

(b)

The fixed payment is given by the par-yield in 2004, calculate as:

$$c = \frac{100 - P_{2004}(10)}{\sum_{t=1}^{10} P_{2004}(t)} = 4.39\%$$

where $P_s(t)$ is the price of zero-coupon bond of maturity t at time s with face value \$100.

Grading:

3 points for the correct answer and formula

2 points for numerical error

Here and below carry-over mistakes are not penalized.

(c)

method A:

You can decompose the interest swap into two legs: a floating note with face value one that pays the floating interest and 100 dollar upon maturity, and a coupon bond that pays the fixed rate as coupon and has a face value of 100. Both legs have maturity 5.

Hence the price of the swap is $P_{2009}(\text{IRS}) = P_{2009}(\text{floating}) - P_{2009}(c)$

The price of the floating note is \$100 since we can generate its cash flows by investing 100 dollars at the going rate and cashing-out the interest rate payment,

The price of the coupon bond is:

$$P_{2009}(c) = 4.38 \cdot \sum_{t=1}^5 P_{2009}(t) + P_{2009}(5) = 111.4$$

Hence, the price of the base IRS is

$$100 - 111.41 = -11.41\$$$

Since the notional amount is 2 billion the contract is worth about -228.2 million dollars.

method B:

Denote by P current(2009) the price of the "old" interest rate swap incepted in 2014. In 2009 there

is a "fresh" IRS (signed in 2009) that matures in 2014, which by definition has price zero. Therefore, the portfolio of longing the old IRS and shorting the new also trades for P.

The cash-flow of this portfolio every period until maturity is $(c' - c)$, where c' is the fixed rate of the "fresh" interest rate swap. The floating rates cancel out. $c' = \frac{100 - P_{2009}(5)}{\sum_{t=1}^5 P_{2009}(t)} = 2.01\%$

Hence, the value of the value of the portfolio is:

$$P = 100 \cdot (c' - c) \cdot \sum_{t=1}^5 P_{2009}(t) = -11.41$$

. Again, accounting for the notional we obtain that the value of the contract is: -228.2 million dollars.

Grading:

6 points for the correct answer

5 points for numerical error/sign error

3 points for using method B but not accounting for the discounting of cash flows

1 point for attempting method b but incorrectly valuing the new fixed rate

0 points otherwise

(d)

Calculate the annually compounded interest:

$$R_{f,2009} = \frac{100}{99.73} - 1 = 0.27\%$$

Grading:

3 points for the correct answer and formula

2 points for numerical error.

(e)

The receiving side of the base IRS gets $100 \cdot (0.27\% - 4.39\%) = -4.12$. Hence, the university will receive a net payment of -82.4 million dollars.

Grading:

3 points for the correct answer and formula

2 points for numerical error.